AGENT IDENTITY & TRUST

PRODUCT CONCEPT PAPER

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Agent Identity & Trust

Product Discovery & Definition

Understanding the identity and trust gap for AI agents operating across digital channels.

Document scope. Discovery identifies the first identity failure to solve. Definition turns that evidence into a channel-facing identity and verification product with a phased architecture and roadmap.

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Individual assignment

DOCUMENT STATUS

Complete concept paper

Discovery 1.1–1.3 · Definition 2.1–2.5

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01 · DISCOVERY

02 · DEFINITION

A focused product decision document for an agent identity and verification platform.

PHASE 1

Discovery

1.1 Context Research

S1 – NIST/NCCoE: Software and AI Agent Identity and Authorization

What it is. A February 2026 draft concept paper from NIST's National Cybersecurity Center of Excellence. It examines identification, authentication, authorisation, delegation and audit for software and AI agents. Its initial scope focuses on enterprise-controlled agents (Booth et al., 2026).
What was found. NIST treats identity as several separate states. A system must define what the Agent ID represents. It must authenticate the active agent, track its credentials, determine its authority and preserve a verifiable audit trail. These states can diverge. An agent may remain registered after its credential expires, its authority changes or its deployment is compromised.
Implication. An Agent Identity Platform cannot be only a directory or one "verified" flag. Persistent identity, principal binding, runtime authentication, delegated authority, lifecycle status and audit history must remain distinguishable. NIST also leaves a wider gap around public-facing and personal agents that cross independently controlled channels.

S2 – OpenID Foundation: Identity Management for Agentic AI

What it is. An October 2025 industry white paper on agent identity, delegated authority, governance and trust across organisational boundaries (OpenID Foundation, 2025).
What was found. Agents often act through a user's identity or session. The receiving service may then see only the user account and cannot tell whether the action came from the person or an autonomous agent. This weakens attribution, least-privilege control, independent revocation and auditability. OpenID argues that the principal and the agent should remain separate actors linked through explicit delegation.
Implication. A relying party needs two linked identities: the principal that granted authority and the agent that performed the action. The evidence must also work outside the system that issued it. Without interoperability each channel would need its own proprietary identity model and trust signal.

S3 – GoDaddy Agent Name Service

What it is. Public 2026 architecture and protocol documentation for an agent registry that uses domain verification, identity certificates and an append-only Transparency Log (GoDaddy, 2026a, 2026b).
What was found. ANS shows that registration and lifecycle transparency are feasible. It can show which domain registered an agent, which version was declared and whether the identity was later changed or revoked. Its own trust model also shows the limit. Domain control does not prove the full legal identity behind an agent. A declared version does not prove that the running code is unchanged. Registration does not show that current behaviour is safe.
Implication. Registration is the foundation of trust, not the final trust decision. A channel still needs current proof that the caller is the registered agent and that its identity or authority has not expired, been suspended or been revoked. Registry status, runtime authentication and contextual validation must remain separate.

Cross-source synthesis

NIST separates the problem into identity, authentication, authority and audit. **OpenID** shows that the agent and principal must remain separate but linked. **GoDaddy ANS** shows that registration does not establish runtime authenticity or current trust.
The common gap is not simply the absence of an Agent ID. Owners, channels and recipients lack a shared and current way to connect a presented agent identity with verified control, runtime authenticity, delegated authority and lifecycle status.

1.2 Stakeholder Insight

Quote.

> The promise of agentic commerce hinges on recognition: the fundamental ability to trust an agent acting on someone's behalf.

Patrick Sullivan, CTO for Security Strategy at Akamai Technologies (Akamai Technologies, 2025)
Stakeholder. Channel Operator. Akamai is a digital infrastructure and security provider positioned between incoming agent traffic and merchant applications. The statement was made in the context of helping merchants accept traffic from autonomous shopping agents while distinguishing legitimate agents from malicious bots.
What it reveals. The obvious need is to recognise whether incoming automation belongs to a legitimate agent. The less obvious need is dual identity. The channel must know which agent is making the request and which person or organisation it represents. A registered Agent ID does not prove that the current caller is that agent or that its authority is still valid. The evidence must be checked during the interaction and combined with current authorisation and risk information before access is granted.
Evidence limitation. This voice strongly supports dual-identity verification in commerce and other high-risk settings. It does not prove that every channel needs the same principal, intent and authorisation information.

1.3 Problem Statement

User. Channel operators that receive AI-agent traffic and protect users, merchants and downstream services.
Goal. Allow legitimate agents to interact with users and protected services while preventing impersonators, compromised agents and unauthorised automation from reaching sensitive systems.
Specific failure. At interaction time the channel cannot reliably verify that an incoming actor is the registered agent it claims to be and that it acts for the claimed principal.
Root cause. Agent identity, principal binding and runtime authentication are fragmented across platform-specific mechanisms. A display name, self-declared agent card or previous registry record does not prove that the active caller controls valid credentials, remains linked to the claimed principal and has not expired, been suspended or been revoked.
Consequence. Channels must either use broad bot controls that block legitimate agents or accept insufficiently verified automation and increase exposure to impersonation, fraud and unauthorised access.

Problem statement

Channel operators trying to allow legitimate AI agents to interact with users and protected services struggle to verify that an incoming actor is the registered agent of its claimed principal because agent identity, principal binding and runtime authentication signals are fragmented and non-interoperable. This results in legitimate agents being blocked or treated as suspicious while impersonators and unauthorised agents may reach users and sensitive systems.

Required answer	Discovery conclusion
Most relevant component	**Authentication.** Registration can show that an identity record exists. It cannot prove that the current caller controls valid credentials for it. Validation and Threat Detection answer later questions about contextual trust and suspicious behaviour.
Critical assumption	**Channel operators are willing and technically able to consume interoperable agent identity evidence and use it in real-time access or trust decisions.** If this is false the first product direction must move away from channel-integrated verification.
Still to validate	Which claims channels need, how evidence changes by interaction risk, who should issue identities and whether verification can meet privacy, latency and implementation constraints.

PHASE 2

Definition

2.1 Key Insights

Observation	Stakeholder	Product decision
Registration creates an agent record, but it does not prove that the actor entering a channel is the same agent or that its credential is still valid.	Channel operator	V1 must include runtime authentication and a channel-facing Verification API. A registry alone is not enough.
A relying party must know both which agent acted and which person or organisation authorised it. Otherwise the agent disappears behind the principal's account.	Channel operator and protected service	Agent and principal remain separate identities linked through verifiable evidence. Runtime proof must preserve both.
A legitimate agent can use an expired credential, operate through the wrong channel, exceed scope or become compromised after registration.	Channel operator, brand and recipient	Identity verification, lifecycle status and action authorisation must remain separate. Trust is not one permanent badge.

2.2 Value Proposition

Value proposition

For **communication channel operators** that need to accept legitimate AI-agent traffic without exposing users and services to impersonation, Agent Trust Gateway is a **registry-backed identity and verification platform** that authenticates agents and links them to verified principals at interaction time. Unlike self-declared metadata, static allowlists and channel-specific bot rules, it provides current and revocable identity evidence that a channel can check before trusting an interaction.

The value proposition addresses the Discovery problem by replacing display names and old registration records with current runtime evidence. It also preserves identity, principal binding, lifecycle status and authority as connected but distinct states (Booth et al., 2026; GoDaddy, 2026a; OpenID Foundation, 2025).

2.3 Product Definition

2.3.1 Product Category

Agent Trust Gateway is a **registry-backed agent identity and trust platform**. It combines persistent identity records, principal verification, credential issuance, runtime authentication, lifecycle

management and a channel-facing verification interface. The pilot is centrally governed so evidence, suspension and revocation rules remain consistent. Federation with approved external issuers is deferred until the shared verification model is proven.

2.3.2 Primary Customer and Scenario

Primary customer: a communication channel operator receiving agent-driven traffic.

First scenario: a brand-owned customer-support agent on a business messaging channel.

Moment	What changes
Before	The channel recognises the business account but not the specific agent generating the conversation. It relies on account-level verification, self-declared metadata and broad bot rules.
Runtime check	The brand registers the agent and binds its Agent ID to the approved messaging account. At conversation start the agent sends a short-lived signed proof. The channel checks Agent ID, principal, channel binding, credential, status and declared purpose.
After	The channel receives a structured result before trusting the interaction. A valid result can produce a narrow recipient-facing signal: **Verified agent identity · Principal verified · Authenticated for this channel**.

First moment of value

The channel can distinguish a legitimate brand agent from unidentified automation during the interaction itself. Invalid, suspended or incorrectly bound identities do not receive the verified result.

2.3.3 Identity Model

An agent identity is a persistent and independently verifiable record for a non-human actor. It identifies the agent separately from its principal and preserves a verified relationship between them.

Record	Core fields	Purpose
Principal Record	Principal ID, type, verified name, assurance level, jurisdiction where required and lifecycle status.	Identifies the person, organisation or system responsible for the agent.
Agent Identity Record	Persistent Agent ID, display name, agent type, linked Principal ID, declared purpose, capabilities, software or model version, public key and lifecycle status.	Represents the independent non-human actor.
Channel Binding	Agent ID, channel type, approved account or endpoint, permitted purpose, status, issue time and expiry.	Lets one Agent ID operate through several verified channels.
Authorisation Grant	Principal ID, Agent ID, capability or action, resource, limits, context, validity window and status.	Records what the agent may do. V1 supports capability scopes. Action-specific receipts come later.

The Agent ID is immutable and does not depend on a display name, phone number, domain or channel account because those values can change or be copied. A software or model update creates a new version under the same Agent ID. A new Agent ID is required when the principal or

control boundary changes enough to represent a different independent actor.

2.3.4 Trust and Security Model

Trust is layered. Registration creates an identity record but does not prove that the current actor is authentic or that every action is permitted.

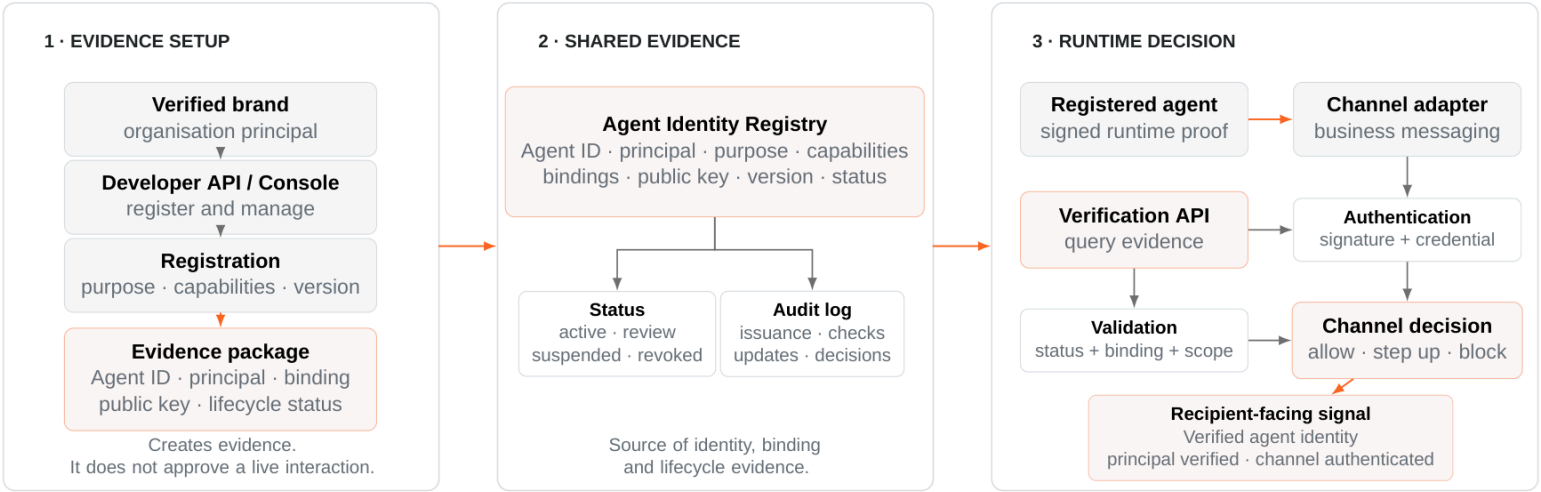
Layer	Question answered	Evidence checked
Establish identity	Who controls the agent before it appears in a channel?	Principal verification, agent registration, channel binding and registered public key.
Verify at runtime	Is the current caller the registered agent?	Short-lived signed proof, credential validity and matching Agent ID.
Apply policy	Should this interaction be trusted now?	Lifecycle status, revocation state, approved channel binding and granted capability scope.

Verification and enforcement sequence

Step	Meaning
1. Verify identity	Principal, Agent ID, channel binding, runtime credential and lifecycle status are valid for this interaction.
2. Apply scope	The signal confirms identity and current control. It does not certify the truth or quality of every response.
3. Step up if sensitive	Refunds, purchases and account changes require a separate action-specific Authorisation Receipt.
4. Investigate abuse	Repeated failures, unapproved channels, reused proofs, unexpected credential changes or verified abuse reports trigger review, suspension or revocation.

2.4 Architectural Schema

V1 is a channel verification path: register evidence once, verify it at interaction time, and step up only for sensitive actions. The schema shows product decisions, not a full service map.



Step-up starts only for sensitive actions such as refunds, purchases or account changes.

PM decision	Product choice
V1 scope	Verify one brand-owned agent on one business messaging channel. V1 includes principal verification, agent registration, channel binding, credential issuance, runtime authentication, validation, lifecycle status, revocation and one channel adapter.
Registry access	Owners update declared metadata. The platform writes assurance, credential and lifecycle status. Channels receive read-only verification access and can submit abuse signals.
Verification model	Verified means principal, Agent ID, channel binding, runtime credential and lifecycle state are valid for the current interaction. It does not certify the truth or quality of every response.
Governance	Before launch, define accepted evidence, assurance levels, suspension authority, appeals, false-positive correction and liability linked to the verified signal.

2.5 Roadmap

Network-effect strategy

The platform starts as a **closed two-sided pilot**, not as an open public registry. One anchor channel agrees to check the verification result, while a small set of high-volume brands supplies the first registered agents. This seeds demand and supply at the same time.

Horizon	Supply side	Demand side	Exit signal
Now 0–3 months	A small set of high-volume brands verifies the organisation principal, registers one production agent, declares purpose and capabilities, binds the Agent ID to one approved messaging account and manages runtime credentials.	One anchor channel integrates the Verification API, checks signed runtime proof, binding, scope and lifecycle status, then returns allow or block with a narrow verified-agent signal.	Pilot agents are recognised reliably. Revoked identities no longer receive a verified result.
Next 3–6 months	More brands onboard through a repeatable flow, manage metadata, rotate credentials and add multiple channel bindings under one Agent ID.	A second channel surface or partner integration checks the same Agent ID. Channels and recipients can report abuse. Step-up approval and short-lived Authorisation Receipts support sensitive actions.	The same Agent ID is verified on more than one surface. Sensitive actions can be approved without broad standing access.
Later 6–12 months	Personal and machine-owned agents, delegation chains and a controlled federation pilot expand supply beyond the first platform.	Messaging, web, voice and API channels reuse evidence. Review and suspension use status and abuse signals without exposing unnecessary private data.	Multiple channels reuse the same evidence and lifecycle events improve decisions without turning trust into one opaque score.

Not building in the first 12 months – universal trust score

The platform will not assign one universal number that claims whether an agent is trustworthy. The first year will not produce enough comparable cross-channel evidence to make that reliable. A single score would also hide the difference between verified identity, valid authorisation and safe behaviour, which could create a false sense of security.

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